Xiaoshan Wu

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EDUCATION

Harvard University — Cambridge, MA, USA
Visiting Scholar (Embodied AI & 4D World Models) — *Apr. 2026 – Present*
– **Advisors:** Yilun Du and Mengyu Wang.

The University of Hong Kong (HKU) — Hong Kong
Ph.D. in Electrical and Computer Engineering — *Sep. 2023 – Present*
– **Awards:** Hong Kong PhD Fellowship (**HKPFS, Top Tier**), HKU Presidential Scholarship.
– **Advisor:** Xiaojuan Qi. **GPA:** 4.0/4.3. **Expected graduation:** Aug. 2027.

Zhejiang University (ZJU) & UIUC — China & USA
Dual B.S. in Computer Engineering (ZJUI Institute) — *Sep. 2019 – Jun. 2023*
– **Honors:** Outstanding Graduate (Rank 5), Dean's List. **GPA:** 3.99/4.0 (ZJU), 3.75/4.0 (UIUC).

RESEARCH INTERESTS

I am interested in **world models for generalizable robotic planning**: how predictions of the physical world can support reliable action in unfamiliar environments. My work develops the perceptual and predictive foundations for this goal, from recovering and retaining spatial evidence to imagining physically plausible 4D futures that guide robot actions and support pre-execution consistency checks. My recent world–action model is evaluated through zero-shot sim-to-real transfer. I am now interested in **agent–world model co-evolution**: using task-relevant prediction errors to guide interaction and model refinement, while action-conditioned world models support policy evaluation, improvement, and longer-horizon planning.

PUBLICATIONS & PREPRINTS

** Equal contribution; † project lead. Published, preprint, and under-review work.*

First- and Co-first-author Work

1. **Verifiable 4D World–Action Model for Robust Robot Control.** [Project]
   X. Wu*, R. Cai*, Y. Bao, Z. Su, A. Wu, F. Wu, Z. Li, Z. Qi, H. Chen, W. Ye, J. Zhang, K. Zhou, M. Wang, and Y. Du.
   Under review at ICLR 2027. *First author*
   S4NDBOX bridges visual imagination and robot control through explicit 4D futures, improving **physical plausibility** (+11.2% relative spatiotemporal IoU vs. RGB-only) and enabling **pre-execution geometric verification** (observed +7.9 pp success within three attempts vs. random ordering). **OOD generalization** over RGB-only: **+13.3 pp** success on RoboTwin Hard (clean-only training); **+40.0** task-score points on single-view **zero-shot sim-to-real** settings.
2. **GaMe: Geometry as Memory for Multi-View Spatial Reasoning.** [Project]
   X. Wu, X. Lyu, Y. Yu, M. Liu, S. Shi, Z. Wang, and X. Qi.
   Under review at ICRA 2027. *First author*
   Geometric working memory aggregates multi-view evidence for spatial question answering.
3. **EAG3R: Event-Augmented 3D Geometry Estimation for Dynamic and Extreme-Lighting Scenes.** [Project]
   X. Wu, Y. Yu, X. Lyu, et al.
   NeurIPS 2025 · Spotlight. *First author*
4. **VideoSSM: Autoregressive Long Video Generation with Hybrid State-Space Memory.** [Project]
   Y. Yu*, **X. Wu***, X. Hu, et al.
   arXiv:2512.04519, 2025. *Co-first author*
5. **LiFR-Seg: Anytime High-Frame-Rate Segmentation via Event-Guided Propagation.** [Project]
   X. Wu, X. Lyu, Y. Yu, B. Wang, Z. Wang, and X. Qi.
   ICLR 2026. *First author*
6. **EVSplat: Uncertainty-Aware Gaussian Splatting for Dynamic Scenes with Event Camera.** [Project]
   Y. Yu*, **X. Wu*†**, X. Lyu, Z. Wang, and X. Qi.
   Under review at AAAI 2027. *Co-first author; project lead*
7. **LiFR v2: Completion-Augmented Event Propagation for High-Rate Dense Prediction.** [Project]
   T. Wan*, **X. Wu*†**, Y. Yu, et al.
   Under review at IEEE TPAMI; arXiv:2609.25803, 2026. *Co-first author; project lead*
8. **Event-Based Depth Prediction With Deep Spiking Neural Network.**
   X. Wu, W. He, M. Yao, et al.
   IEEE TCDS, 16(6):2008–2018, 2024. *First author*

Collaborative Work

9. **Efficient and Accurate Neural-Field Reconstruction Using Resistive Memory.**
   Y. Yu, X. Zhang, S. Wang, . . . , **X. Wu**, et al. **Nature**, 654:642–651, 2026.
10. **OptiGeo: Efficient Monocular Geometry for Embodied Perception in Optically Challenging Scenes.**
    M. Liu, T. Liu, J. Xia, X. Lyu, **X. Wu**, et al. **CoRL 2026**.
11. **FoundationGeo: Learning Spatial Pixel-Wise Fields for Monocular Metric Geometry.**
    M. Liu, X. Lyu, T. Ren, P. Dai, **X. Wu**, et al. **ECCV 2026**.
12. **Stabilizing Streaming Video Geometry via Dynamic Feature Normalization.**
    X. Lyu, M. Liu, **X. Wu**, et al. **CVPR 2026**, pp. 7577–7587.
13. **Continuous-Time Digital Twin with Analog Memristive Neural Ordinary Differential Equation Solver.**
    H. Chen, J. Yang, J. Chen, . . . , **X. Wu**, et al. **Science Advances**, 11(22):eadr7571, 2025.
14. **Topology Optimization of Random Memristors for Input-Aware Dynamic SNN.**
    B. Wang, X. Zhang, S. Wang, . . . , **X. Wu**, et al. **Science Advances**, 11(16):eads5340, 2025.
15. **Charge-Selective 2D Heterointerface-Driven Multifunctional Floating Gate Memory for In Situ Sensing-Memory-Computing.**
    C. Li, X. Chen, Z. Zhang, **X. Wu**, et al. **Nano Letters**, 24(47):15025–15034, 2024.

TECHNICAL SKILLS

Tools: Python, PyTorch (CUDA), Diffusers, Open3D, OpenCV.
Methods: World–action modeling; geometric action checks; sim-to-real evaluation on Franka Research 3.

SELECTED AWARDS

Hong Kong PhD Fellowship (HKPFS, 2023); HKU Presidential Scholarship (2023).
Outstanding Graduate of Zhejiang University (2023); NeurIPS (2025) and ICLR (2026) Travel Awards.